

高硬钢用4刃长颈平底铣刀
Hardened Steels 4-Flute Long Neck Square Endmills

4刃·平底铣刀

4-Flute·Square

平面

侧切

槽切

粗加工

半精加工

精加工

切削条件

外径公差

1≤D≤3: 0~-0.01
4≤D: 0~-0.015

h5

0~-0.005

(mm)

- 碳素钢

Carbon Steel

P
- 合金钢

Alloy Steel

P
- 调质钢

Prehardened Steel

P
- 高硬度钢

Hardened Steel

H
- 高硬度钢

Hardened Steel

H
- 高硬度钢

Hardened Steel

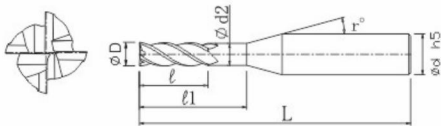
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- 在 高硬度领域 (48~65HRC) 的长时间加工中也可实现稳定的长寿命。
- 改进了刀刃形状，改善了倾斜量和加工精度。
- Realized stably long tool life against high hardened steels (48 ~ 65HRC).
- Improved milling deflection and accuracy by advanced flute design.

产品代码 Code No.	规格型号 Spec Typ.	外径(D) Dia.	颈长(l1) Under Neck Length	刃长(l) Length of Cut	颈径(d2) Neck Dia	颈角(r) Neck Taper Angle	柄径(d) Shank Dia.	全长(L) Overall Length	定价 Retail Price
FX6MNE1004	1x4L	1	4	0.8	0.95	15°	4	50	
FX6MNE1006	1x6L		6	0.8	0.95	15°	4	50	
FX6MNE1008	1x8L		8	0.8	0.95	15°	4	50	
FX6MNE1010	1x10L		10	0.8	0.95	15°	4	50	
FX6MNE1012	1x12L		12	0.8	0.95	15°	4	50	
FX6MNE1016	1x16L		16	0.8	0.95	15°	4	50	
FX6MNE1506	1.5x6L	1.5	6	1.2	1.45	15°	4	50	
FX6MNE1508	1.5x8L		8	1.2	1.45	15°	4	50	
FX6MNE1510	1.5x10L		10	1.2	1.45	15°	4	50	
FX6MNE1512	1.5x12L		12	1.2	1.45	15°	4	50	
FX6MNE1516	1.5x16L		16	1.2	1.45	15°	4	50	
FX6MNE2008	2x8L	2	8	1.6	1.94	15°	4	50	
FX6MNE2010	2x10L		10	1.6	1.94	15°	4	50	
FX6MNE2012	2x12L		12	1.6	1.94	15°	4	50	
FX6MNE2016	2x16L		16	1.6	1.94	15°	4	50	
FX6MNE2018	2x18L		18	1.6	1.94	15°	4	50	
FX6MNE2020	2x20L		20	1.6	1.94	15°	4	50	
FX6MNE3010	3x10L	3	10	4.5	2.85	15°	4	50	
FX6MNE3012	3x12L		12	4.5	2.85	15°	4	50	
FX6MNE3016	3x16L		16	4.5	2.85	15°	4	50	
FX6MNE3020	3x20L		20	4.5	2.85	15°	4	50	
FX6MNE3010L	3x10L		8	4.5	2.85	15°	6	60	
FX6MNE3012L	3x12L		12	4.5	2.85	15°	6	60	
FX6MNE3016L	3x16L	4	16	4.5	2.85	15°	6	60	
FX6MNE3020L	3x20L		20	4.5	2.85	15°	6	60	
FX6MNE3025L	3x25L		25	4.5	2.85	15°	6	75	
FX6MNE3030L	3x30L		30	4.5	2.85	15°	6	75	
FX6MNE4012L	4x12L		12	6	3.8	15°	6	60	
FX6MNE4016L	4x16L		16	6	3.8	15°	6	60	
FX6MNE4020L	4x20L	4	20	6	3.8	15°	6	60	
FX6MNE4025L	4x25L		25	6	3.8	15°	6	75	
FX6MNE4030L	4x30L		30	6	3.8	15°	6	75	

高硬度材料加工用。超群的切削排出性能，可实现高效率加工。
For machining of high-hardness materials. Excellent chip removal performance enables high efficiency.

TYPE A:



切削参数参考表
Recommended Milling Conditions

加工材料 Work Material		碳素钢·合金钢·不锈钢 Carbon Steels·Alloy Steels·Stainless Steels S50C·SCM·SKD·SUS				调质钢 Prehardened Steels NAK55·NAK88·HPM-1 (~43HRC)			
		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
外 径 Dia.	颈长(l1) Under Neck Length	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
1	4	25,000	1,700	0.055	0.6	22,000	1,100	0.045	0.6
	6	20,000	1,200	0.045	0.6	18,000	750	0.035	0.6
	8	18,000	1,050	0.035	0.6	15,000	600	0.025	0.6
	10	16,000	900	0.025	0.6	14,000	520	0.018	0.6
	12	14,000	750	0.02	0.6	12,000	450	0.014	0.6
1.5	6	20,000	1,800	0.08	0.9	16,000	1,100	0.07	0.9
	8	18,000	1,500	0.07	0.9	14,000	900	0.06	0.9
	10	16,000	1,300	0.06	0.9	13,000	780	0.05	0.9
	12	14,000	1,050	0.05	0.9	12,000	670	0.04	0.9
	16	12,000	750	0.035	0.9	10,000	480	0.025	0.9
2	8	16,000	2,000	0.11	1.2	13,000	1,200	0.09	1.2
	10	16,000	1,800	0.1	1.2	13,000	1,100	0.08	1.2
	12	14,000	1,500	0.09	1.2	12,000	1,000	0.07	1.2
	16	12,000	1,200	0.07	1.2	10,000	750	0.05	1.2
	18	11,000	1,000	0.06	1.2	9,000	650	0.04	1.2
3	20	10,000	900	0.05	1.2	8,000	550	0.035	1.2
	8	11,000	2,300	0.2	1.8	9,000	1,500	0.16	1.8
	10	11,000	2,200	0.18	1.8	9,000	1,300	0.14	1.8
	12	11,000	2,000	0.16	1.8	9,000	1,200	0.12	1.8
	16	10,000	1,500	0.12	1.8	8,000	900	0.09	1.8
4	20	9,000	1,200	0.1	1.8	7,000	680	0.07	1.8
	25	8,000	1,050	0.08	1.8	6,000	570	0.06	1.8
	30	7,000	900	0.06	1.8	5,000	450	0.045	1.8
	12	8,000	2,200	0.3	2.4	6,000	1,300	0.25	2.4
	16	8,000	2,100	0.25	2.4	6,000	1,200	0.2	2.4
4	20	8,000	2,000	0.2	2.4	6,000	1,100	0.15	2.4
	25	7,000	1,700	0.15	2.4	5,000	900	0.1	2.4
	30	7,000	1,500	0.12	2.4	5,000	800	0.08	2.4
备注 Notes		※ 1切削合金钢、不锈钢时，请将主轴转速及进给速度降低至80%时的值作为参考值。 ※ 1 Reference value for Alloy and Stainless Steels are 80% of recommended cutting conditions.							
		※ 本切削参数仅供参考。请根据实际的加工形状及使用机床等调整切削参数。 ※ 切深量的ap表示轴向切入量。ae表示距量 ※ 请在考虑加工材料及加工形状等的基础上，选用合适的切削油。 ※ 加工深沟时，请充分注意切削油的供油及排屑是否顺畅。 ※ 轴向进刀建议采用螺旋进刀及倾斜进刀方式。 ※ 由于立面附近的刀具负荷会增大，必须降低进给速度和切深量。 ※ 沟槽切削时建议将切削参数表的切深量降低至80%时的值作为参考值，并采用来回切削加工方式。l1(颈长)/D(刃径)为5以上时，建议先使用有效长较短的刀具切削初期定位槽。 ※ 发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率调整。 ※ 刀具的伸出量超过柄径的5倍时，受铣刀刀柄跳动精度等的影响，有时必须对加工参数进行大幅调整。此时请按相同的比率降低主轴转速和进给速度等，调整切削参数。							
		※ These recommended cutting conditions indicate just reference. It should be adjusted according to milling shape and machine type. ※ ap:Axial Depth of Cut, ae:Radial Depth of Cut. ※ Select a cutting fluid appropriate to work material, milling shape and machining content. ※ Coolant supply and chip disposal in the deep portion are very important. ※ Recommend to apply herical or ramping for approaching into axial direction. ※ Reduction of feed and Depth of Cut to reduce machining load around side wall. ※ When grooving cutting, it is recommended to reduce the cutting depth of the cutting parameter table to 80% as a reference value, and use the back and forth cutting mode. When l1(Under Neck Length)/D(blade diameter) is more than 5, it is recommended to use a tool with a shorter effective length to cut the initial positioning slot. ※ Reduce both spindle speed and feed at same rate for chattering and also for insu cient spindle speed of a machine. ※ Major adjustment of milling conditions, e.g. adjust spindle and feed speed at same rate, required on condition of a tool overhang length exceeding a shank diameter 5 times due to possible accuracy impact by chuck runout etc.							

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